

ECON 100: Introduction to Economics

Exam 1, Fall 2026

- Define any variables and/or state any assumptions you need to answer the problems.
 - All materials should be put away before beginning the exam. Use of cell phones during the exam for any purpose is forbidden. The Honor Code applies to this exam.
 - Please write your answers in the space provided.
 - Keep your answers short but clear. Your goal is to convince a skeptical grader that you understand the relevant concepts well enough to answer the question you are given.
 - The questions on the exam sum to 60 points.
 - Good luck on your Econ test!
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1. (2 points) What is your name?
2. (2 points) What section are you in?

1 True/False Questions

Indicate “T”rue or “F”alse for each of the following statements or claims. For each false statement, if you correctly and clearly explain why it is false, I will give you a bonus point. No explanations necessary for claims or statements. which you think are true.

3. (2 points) Suppose there are two firms whose PPF includes the points (0,100) and (100,0). Firm A has constant opportunity cost between these two points, and Firm B has increasing opportunity cost between these two points. **Claim:** Firm A has more feasible bundles than Firm B.
4. (2 points) Suppose Wanda calculates one Dunkin Donuts coffee will increase her utility by \$6 worth of utils, compared to having 0 Dunkin Donuts coffees. To produce this additional coffee costs Dunkin Donuts an additional \$4. **Claim:** A mutually beneficial trade would be Wanda handing Dunkin Donuts \$6 in exchange for a coffee.
5. (2 points) Andrew and Anna need to hand-make 20 fliers and 4 posters to advertise for the first home volleyball game of the season in order to post around campus. Andrew and Anna have both taken Econ 100. Andrew makes the 4 posters, and Anna makes the 20 fliers. **Claim:** Based on what we have learned in class, this suggests Anna has the comparative advantage in making fliers, and Andrew has the comparative advantage in making posters.
6. (2 points) Suppose that Wegman’s starts labeling the exact dairy farm which produced each gallon of 2% milk they sell in their grocery store, and organizing the shelves by farm. Seeing this, Farmer Dan decides that he should increase the price of his 2% milk by \$0.10 at Wegman’s since he believes his milk is better than the other farmers who sell their milk at Wegman’s. After the price increase, Farmer Dan’s milk sales at Wegman’s fall from 100 per week to 0 per week. **Claim:** Consumer demand for Farmer Dan’s 2% milk is perfectly elastic.

7. (2 points) **Claim:** As the number of buyers and sellers increases, both the market demand and the market supply curves become less responsive to changes in price.

2 Multiple Choice Questions

Circle the best answer to each question. There is only one answer for each question. No explanation necessary.

8. Suppose you are reviewing Dana's grocery receipts from the past month. You focus on her purchases of green onions and tofu. You calculate her price elasticity of quantity demanded for tofu at -0.5 , her income elasticity of quantity demanded for green onions at 0.75 , and her cross price elasticity of quantity demanded for tofu and green onion at 1.4 .
- (a) (2 points) Using only the information above, which of the following statements **must be true** based on these calculated elasticities?
- A. Tofu is an inferior good for Dana.
 - B. Green onions and tofu are substitutes for Dana.
 - C. Green onions are an ordinary good for Dana.
 - D. Green onions and tofu are complements for Dana.
- (b) (2 points) Using only the information above, which of the following statements **must be false** based on these calculated elasticities.
- A. Tofu is an ordinary good for Dana.
 - B. Tofu and green onions are both ordinary goods for Dana.
 - C. Green onions are an inferior good for Dana.
 - D. Dana's demand for tofu and green onions is elastic.
9. (2 points) Bill (from HW2) is back at the same arcade after renovations, but Bill can only stay for 1 hour (60 minutes). Both pinball and the basketball game are now free. Each pinball game takes Bill 4 minutes to play, and each basketball game takes Bill 6 minutes to play. **Question:** Which of the following represents an efficient bundle of pinball and basketball games for Bill?
- A. 0 pinball games, 12 basketball games.
 - B. 2 pinball games, 9 basketball games.
 - C. 10 pinball games, 0 basketball games.
 - D. 3 pinball game, 8 basketball games.

10. (2 points) The Table below represents Avery's total costs and total revenue from giving a nail trim to a given quantity of cats.

# Cats	Total Revenue	Total Costs
0	0	2
1	6	8
2	14	16
3	26	26
4	32	40

Question: What is Avery's profit-maximizing quantity of cat nail trims?

- A. 1 cat.
 - B. 2 cats.
 - C. 3 cats.
 - D. 4 cats.
11. (2 points) Suppose demand is given by $Q_D = 100 - 5P$. Using the midpoint formula, calculate the price elasticity of quantity demanded between when $P = 4$ and $P = 12$.
- A. $-\frac{2}{3}$
 - B. $-\frac{3}{2}$
 - C. -1.
 - D. $-\frac{1}{2}$

3 Short Answer Questions

These questions all require an explanation. Remember you are trying to convince me you understand the why and the how of what you are doing, not simply getting the answer correct. Cite specific concepts from class in your answers for full credit.

12. **The Merger of Pawnee and Eagleton.** Pawnee and Eagleton are two towns which have public parks that require regular maintenance and what we will call other governmental services. In Pawnee, every government worker can maintain 2 parks or perform 1 other governmental service. In Eagleton, every government worker can maintain 1 parks or perform 1 other governmental service. Each town has 50 government workers.

(a) (8 points) Draw the PPF for each town on separate graphs. Be sure to label your axes and which graph corresponds to which town.

(b) (6 points) Which town has the absolute advantage in maintaining public parks. Which town has the absolute advantage in performing other governmental services? Briefly explain your answer.

(c) (6 points) Which town has the comparative advantage in maintaining public parks? Which town has the comparative advantage in performing other governmental services? Briefly explain your answers.

- (d) (4 points) Suppose that both towns currently use half of their workers to maintain public parks and the other half to perform other governmental services. How much of each service does each town provide? No explanation necessary.
- (e) (6 points) Suppose Pawnee and Eagleton are located next to each other, and allow workers to work for either government. Propose a trade regarding number of parks maintained and other governmental services provided that would be mutually beneficial to both towns. As part of your answer, explain why the price of this trade would be acceptable to both towns.
- (f) (6 points) Suppose instead Pawnee and Eagleton decide to merge. Draw a new PPF of this newly combined town and explain the shape of this combined PPF.

Extra Scrap Page